# Overview of Likelihood Training Using Forward-Backwards SDE Theory

Let $\Omega = \mathcal{C}([0,T],\mathbb{R}^d)$ be the path space of all continuous functions describing trajectories from time $t=0$ to $t=T$, and denote $\mathcal{P}(\Omega)$ the space of probability measures on $\Omega$. Consider $P \in \mathcal{P}(\Omega)$ a reference measure defined as the law of the continuous-time process $(X_t)_{t\in[0,T]}$ governed by the following SDE:

$$\mathrm{d}X_t = f(t,X_t)\mathrm{d}t + g(t)\mathrm{d}W_t$$

where $f : [0,T]\times\mathbb{R}^d \to \mathbb{R}^d$ is the drift function, $g(t) : [0,T] \to \mathbb{R}$ the diffusion coefficient, and $(W_t)_{t\in[0,T]}$ the standard $n$-dimensional Wiener process.

Let $p_{\text{data}}, p_{\text{prior}} \in \mathcal{P}(\mathbb{R}^d)$ be the starting ($t=0$) and ending ($t=T$) marginals, respectively, and let $\mathcal{P}(p_{\text{data}}, p_{\text{prior}}) \subset \mathcal{P}(\Omega)$ denote the space of all admissible path measures whose time marginals satisfy those endpoints. The Schrödinger Bridge problem can then be formulated as:

$$\min_{Q\in\mathcal{P}(p_{\text{data}},p_{\text{prior}})} \mathrm{KL}(Q \mid P) \tag{S}$$

Let $\Psi, \widehat{\Psi} : [0,T]\times\mathbb{R}^d \to \mathbb{R}$ be the solutions to the followinbg system of PDEs:

$$\begin{aligned}
\frac{\partial\Psi}{\partial t} &= -\nabla_x\Psi^\top f - \frac{1}{2}\mathrm{Tr}(g^2\nabla_x^2\Psi)\\
\frac{\partial\widehat{\Psi}}{\partial t} &= -\nabla_x\cdot(\widehat{\Psi}f) + \frac{1}{2}\mathrm{Tr}(g^2\nabla_x^2\widehat{\Psi})
\end{aligned}$$

that are coupled together from the initial and final points:

$$\Psi(0,\cdot)\widehat{\Psi}(0,\cdot) = p_{\text{data}}(\cdot), \qquad \Psi(T,\cdot)\widehat{\Psi}(T,\cdot) = p_{\text{prior}}(\cdot)$$

Then the solution to (S) can be expressed by the path measure of the following forward (or equivalent backward) SDE:

$$\mathrm{d}X_t = [f(t,X_t) + g^2(t)\nabla_x\log\Psi(t,X_t)]\mathrm{d}t + g(t)\mathrm{d}W_t, \quad X_0 \sim p_{\text{data}} \tag{1a}$$

$$\mathrm{d}X_t = [f(t,X_t) - g^2(t)\nabla_x\log\widehat{\Psi}(t,X_t)]\mathrm{d}t + g(t)\mathrm{d}W_t, \quad X_T \sim p_{\text{prior}} \tag{1b}$$

where $\nabla_x\log\Psi(t,X_t)$ and $\nabla_x\log\widehat{\Psi}(t,X_t)$ are the optimal forward and backward drifts of the Schrödinger Bridge. It also holds that for every $t\in[0,T]$:

$$p_t^{\text{SB}}(X_t) = \Psi(t,X_t)\widehat{\Psi}(t,X_t)$$

The exact proof can be found in Likelihood Training of Schrödinger Bridge Using Forward-Backward SDEs Theory.

Consider the following coupled SDEs:

$$\mathrm{d}X_t = f(t, X_t)\mathrm{d}t + G(t, X_t)\mathrm{d}W_t, \quad X_0 = x_0 \tag{2a}$$
$$\mathrm{d}Y_t = -h(t, X_t, Y_t, Z_t)\mathrm{d}t + Z_t^\top \mathrm{d}W_t, \quad Y_T = \varphi(X_T) \tag{2b}$$

where the functions $f$, $G$, $h$ and $\phi$ satisfy proper regularity conditions so that there exists a pair of unique solutions satisfying 2a and 2b.

Now consider the following second order parabolic PDE and suppose $u(t, x)$ is once continuously differentiable in $t$ and twice in $x$:

$$\frac{\partial u}{\partial t} + \frac{1}{2}\mathrm{Tr}(\nabla_x^2 u G G^\top) + (\nabla_x u)^\top f + h(t, x, u, G^\top \nabla_x u) = 0, \quad u(T, x) = \phi(x) \tag{3}$$

Then the solution to 2a and 2b coincides with the solution to 3 among paths generated by the foward 2a, almost surely, meaning that the following stichastic representation (knows as the non-linear Feynman-Kac relation) is valid:

$$u(t, X_t) = Y_t, \qquad G(t, X_t)^\top \nabla_x u(t, X_t) = Z_t$$

It is knows that the deterministic PDE solution $u(t, x)$ can be recovered by taking conditional expectation:

$$u(t, x) = \mathbb{E}_{X_t \sim 2a}[Y_t \mid X_t = x], \quad G(t, x)^\top \nabla_x u(t, x) = \mathbb{E}_{X_t \sim 2a}[Z_t \mid X_t = x]$$

This result provides the essential mathematical translation required to convert intractable PDEs into computationally solvable SDEs. The optimality conditions for the Schrödinger Bridge are defined by two coupled partial differential equations (PDEs) that are notoriously difficult to solve. This result establishes that solutions to a specific class of nonlinear PDEs can be exactly represented by the solutions to a Forward-Backward SDE (FBSDE) system.

Now, consider the following set of coupled SDEs:

$$\mathrm{d}X_t = (f(t, X_t) + g(t)Z_t)\mathrm{d}t + g(t)\mathrm{d}W_t \tag{4a}$$
$$\mathrm{d}Y_t = \frac{1}{2}Z_t^\top Z_t \mathrm{d}t + Z_t^\top \mathrm{d}W_t \tag{4b}$$
$$\mathrm{d}\widehat{Y}_t = \left[\frac{1}{2}\widehat{Z}_t^\top \widehat{Z}_t + \nabla_x \cdot (g(t)\widehat{Z}_t - f(t, X_t)) + \widehat{Z}_t^\top Z_t\right]\mathrm{d}t + \widehat{Z}_t^\top \mathrm{d}W_t \tag{4c}$$

where the boundary conditions are given by:

$$X_0 = x_0, \qquad Y_T + \widehat{Y}_T = \log p_{\text{prior}}(X_T)$$

Supose $\Psi$, $\widehat{\Psi}$ are once continuously differentiable in $t$ and twice in $x$, then the non-linear Feynman-Kac relations between equations 4 and the initial PDEs of the Schrödinger problem S are given by:

$$Y_t \equiv Y(t, X_t) = \log \Psi(t, X_t), \quad Z_t \equiv Z(t, X_t) = g(t)\nabla_x \log \Psi(t, X_t)$$
$$\widehat{Y}_t \equiv \widehat{Y}(t, X_t) = \log \widehat{\Psi}(t, X_t), \quad \widehat{Z}_t \equiv \widehat{Z}(t, X_t) = g(t)\nabla_x \log \widehat{\Psi}(t, X_t)$$

Furthermore, $Y_t$, $\widehat{Y}_t$ obey the following relation:

$$Y_t + \widehat{Y}_t = \log p_t^{\mathrm{SB}}(X_t)$$

Given the solution of the FBSDE system 4 ($Z_t$, $\widehat{Z}_t$) the log-likelihood of the SB model at a data point $x_0$ can be expressed as:

$$\log p_0^{\mathrm{SB}}(x_0) = \mathbb{E}[\log p_T(X_T)] - \int_0^T \mathbb{E}\left[\frac{1}{2}\|Z_t\|^2 + \frac{1}{2}\|\widehat{Z}_t\|^2 + \nabla_x \cdot (g\widehat{Z}_t - f) + \widehat{Z}_t Z_t\right] \mathrm{d}t \tag{5}$$

where the expectation is taken over the forward 4a with initial condition $X_0 = x_0$.

Now we have everything we need to craft the learning algorithm. Let's summarize one last time what we have done. We started with the two PDEs which provide the potentials $\Psi$ and $\widehat{\Psi}$ of the Schrödinger Bridge (SB). If we had access to them, we could simply simulate (1a) or (1b) in order to sample from the SB. But unfortunately, these PDEs are computationally intractable to solve directly for high-dimensional generative applications.

To resolve this intractability, the authors use the nonlinear Feynman-Kac relation on the initial PDEs to derive the Forward-Backward SDEs (FBSDEs): (4a), (4b), and (4c). This stochastic representation allows us to write equation (5), which we will now use as the divergence-based log-likelihood objective to train our parameterized networks.

We define the log-likelihood loss as:

$$\begin{aligned} L_{\mathrm{SB}}(x_0;\ \theta, \phi) = \mathbb{E}_{X_T \sim (4a)}[\log p_{\mathrm{prior}}(X_T)] \\ - \int_0^T \mathbb{E}_{X_t \sim (4a)}\Big[\frac{1}{2}\|Z(t, X_t; \theta)\|^2 + \frac{1}{2}\|\widehat{Z}(t, X_t; \phi)\|^2 \\ + \nabla_x \cdot (g(t)\widehat{Z}(t, X_t; \phi) - f(t, X_t)) \\ + \widehat{Z}(t, X_t; \phi)^\top Z(t, X_t; \theta)\Big] \mathrm{d}t \end{aligned} \tag{6}$$

Crucially, the authors established that this expression forms a parameterized lower bound to the true log-likelihood of the data, meaning $\log p_0^{\mathrm{SB}}(x_0) \geq L_{\mathrm{SB}}(x_0;\ \theta, \phi)$.

Therefore, the objective is to maximize this lower bound w.r.t both $\theta$ and $\phi$. This joint maximization acts as Maximum Likelihood Estimation (MLE), training both the forward and backward policies (the potentials $\Psi$, $\widehat{\Psi}$) simultaneously to push the model's generated distribution to match the true data distribution.

By isolating the terms in (6) that depend on $\phi$, the maximization over $\phi$ yields:

$$\max_{\phi} \tilde{L}_{\mathrm{SB}}(x_0;\ \phi) = -\int_0^T \mathbb{E}_{X_t\sim(4a)}\Big[\frac{1}{2}\|\widehat{Z}(t,X_t;\phi)\|^2 + g(t)\nabla_x \cdot \widehat{Z}(t,X_t;\phi) + \widehat{Z}(t,X_t;\phi)^\top Z(t,X_t;\theta)\Big]\mathrm{d}t \tag{7}$$

To jointly optimize the forward policy $\theta$ using the forward SDE (4a), the framework must perform backpropagation through time across the entire sequential simulation. Because $\theta$ dictates the step-by-step state transitions, storing this sequential computational graph is prohibitively expensive for high-dimensional image networks. Conversely, updating the backward policy $\phi$ on this forward pass is memory-efficient because it merely evaluates the fixed generated states without governing the trajectory itself.

To bypass the memory bottleneck for $\theta$, the authors discard the computational graph entirely during the forward simulation and instead leverage the symmetric structure of the Schrödinger Bridge. By simulating backwards from the prior $X_T \sim p_{\mathrm{prior}}$ (also discarding its computational graph), they can evaluate and update $\theta$ against newly fixed backward states. Specifically, these backward trajectories are sampled by substituting the current backward policy network into (1b), yielding the following parameterized backward SDE:

$$\mathrm{d}X_t = (f(t,X_t) - g(t)\widehat{Z}(t,X_t;\phi))\mathrm{d}t + g(t)\mathrm{d}W_t \tag{8}$$

Using these sampled paths, the alternating maximization objective for the forward policy is precisely defined as:

$$\max_{\theta} \tilde{L}_{\mathrm{SB}}(X_T;\ \theta) = -\int_0^T \mathbb{E}_{X_t\sim(8)}\Big[\frac{1}{2}\|Z(t,X_t;\theta)\|^2 + g(t)\nabla_x \cdot Z(t,X_t;\theta) + \widehat{Z}(t,X_t;\phi)^\top Z(t,X_t;\theta)\Big]\mathrm{d}t \tag{9}$$

It is important to note that the authors do not provide a strict mathematical proof guaranteeing convergence for these specific divergence-based objectives. Formal convergence analysis is explicitly left as future work. However, the authors emphasize that this alternating training scheme mechanically mirrors the classical Iterative Proportional Fitting (IPF) algorithm used in traditional Schrödinger Bridge problems. Because classical IPF has well-established theoretical convergence guarantees, the authors conjecture that a similar convergence analysis can be extended to their continuous-time SDE framework, relying on this conceptual alignment alongside their empirical results.

**Algorithm 1:** Likelihood Training of SB-FBSDEs

**Input:** Boundary distributions $p_{\text{data}}$, $p_{\text{prior}}$, parameterized policies $Z(\cdot, \cdot;\ \theta)$, $Z(\cdot, \cdot;\ \phi)$

**while** *Not Converged* **do**
- **for** $k = 1$ **to** $K$ **do**
  - **if** $k\%M == 0$ **then**
    - Sample $(X_t)_{[0,T]}$ from 4a with $X_0 \sim p_{\text{data}}$
  - Compute $\tilde{L}_{\text{SB}}(x_0; \phi)$.
  - Update $\phi$ with $\nabla_\phi \tilde{L}_{\text{SB}}(x_0; \phi)$
- **for** $k = 1$ **to** $K$ **do**
  - **if** $k\%M == 0$ **then**
    - Sample $(X_t)_{[0,T]}$ from 8 where $X_T \sim p_{\text{prior}}$
  - Compute $\tilde{L}_{\text{SB}}(X_T; \theta)$
  - Update $\theta$ with $\nabla_\theta \tilde{L}_{\text{SB}}(X_T; \theta)$

**Output:** Trained policies $Z^*(\cdot, \cdot;\ \theta)$, $Z^*(\cdot, \cdot;\ \phi)$

# Algorithmic Implementation

To generate new data samples after training, one could simply simulate the parameterized backward SDE (8) from $t = T$ to $t = 0$. However, the propagation of SDEs via numerical methods like the Euler-Maruyama method introduces discretization errors that accumulate over time. To mitigate this, the authors implement a predictor-corrector sampling procedure inspired by Score-Based Generative Models.

The procedure alternates between two steps at each time interval:

1. Predictor Step: The state is advanced numerically using the trained backward policy $\widehat{Z}_\phi$.
2. Corrector Step: The algorithm applies Langevin dynamics to refine the sample distribution at each time step.

Langevin dynamics require the marginal score $\nabla_x \log p_t^{\text{SB}}$ to push the samples toward the true optimal density. Because the Schrödinger Bridge mathematically guarantees that $Z_t + \widehat{Z}_t = g(t)\nabla_x \log p_t^{\text{SB}}$, the authors can estimate this score directly using the sum of the two trained networks:

$$\nabla_x \log p_t^{\text{SB}} \approx \frac{1}{g(t)} \left[ Z(t, X_t; \theta) + \widehat{Z}(t, X_t; \phi) \right]$$

Below is the generation algorithm, which uses a pre-defined signal-to-noise ratio $r$ to scale the Langevin noise $\sigma_{t,i}$ appropriately during the corrector steps. To strictly preserve the reference measure of the Schrödinger Bridge, the predictor step is formulated as the exact Euler-Maruyama discretization of the general backward SDE (1b).

**Algorithm 2:** Generative Process of SB-FBSDE (Predictor-Corrector)

**Input:** Prior $p_{\text{prior}}$, trained policies $Z(\cdot,\cdot;\,\theta)$, $\widehat{Z}(\cdot,\cdot;\,\phi)$, signal-to-noise ratio $r$, corrector steps $N$

Sample $X_T \sim p_{\text{prior}}$;
**for** $t = T$ **to** $\Delta t$ *step* $-\Delta t$ **do**
  Sample $\epsilon \sim \mathcal{N}(0, I)$;
  Predict Step:
  $X_{t,1} \leftarrow X_t - \left[f(t, X_t) - g(t)\widehat{Z}(t, X_t; \phi)\right]\Delta t + g(t)\sqrt{\Delta t}\epsilon$;
  **for** $i = 1$ **to** $N$ **do**
    Sample $\epsilon_i \sim \mathcal{N}(0, I)$;
    Estimate Score using both policies:
    $S(X_{t,i}) = \frac{1}{g(t)}\left[Z(t, X_{t,i}; \theta) + \widehat{Z}(t, X_{t,i}; \phi)\right]$;
    Compute Noise Scale:
    $\sigma_{t,i} = \dfrac{2r^2 g(t)^2 \|\epsilon_i\|^2}{\|Z(t, X_{t,i}; \theta) + \widehat{Z}(t, X_{t,i}; \phi)\|^2}$;
    Correct Step:
    $X_{t,i+1} \leftarrow X_{t,i} + \sigma_{t,i} S(X_{t,i}) + \sqrt{2\sigma_{t,i}}\epsilon_i$;
  Propagate $X_{t-\Delta t} \leftarrow X_{t,N}$;
**Output:** Generated data $X_0$

While the algorithms above outline the core mathematical framework, a few practical details are required to implement and evaluate SB-FBSDEs efficiently:

1. Divergence Estimation: The exact computation of the divergence terms $\nabla_x \cdot Z_\theta$ and $\nabla_x \cdot \widehat{Z}_\phi$ in the training objectives (7) and (9) requires evaluating the trace of the network's Jacobian. This is computationally prohibitive for high-dimensional data like images. To resolve this, the paper notes that the divergence must be efficiently approximated using Hutchinson's trace estimator..

2. Probability Flow ODE for Evaluation: Simulating SDEs provides generated samples, but calculating exact Negative Log-Likelihoods (NLL) on a test set requires a deterministic mapping. To achieve this, the authors prove that the Schrödinger Bridge admits a Probability Flow ODE that shares the exact same marginal den-

sities as the optimal SDEs. This ODE is defined as:

$$\mathrm{d}X_t = \left[ f(t, X_t) + g(t) Z(t, X_t; \theta) - \frac{1}{2} g(t) \left( Z(t, X_t; \theta) + \widehat{Z}(t, X_t; \phi) \right) \right] \mathrm{d}t$$

By integrating this ODE backward in time, one can use standard continuous normalizing flow techniques to compute the exact NLL of the model during evaluation.